Sri Lanka Institute of Information Technology

Network Design & Management (IT1030)

Lab Report 02

Installing and configuring DHCP

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Batch: IT_WD_03.02

Introduction

DHCP means that Dynamic Host Configuration Protocols. The main job of DHCP is assigning separate and unique Ip addresses and other configurations for each device automatically. DHCP server configures in CentOS and checks on Fedora. DHCP use UDP and TCP/Ip protocols TCP is better than UDP.

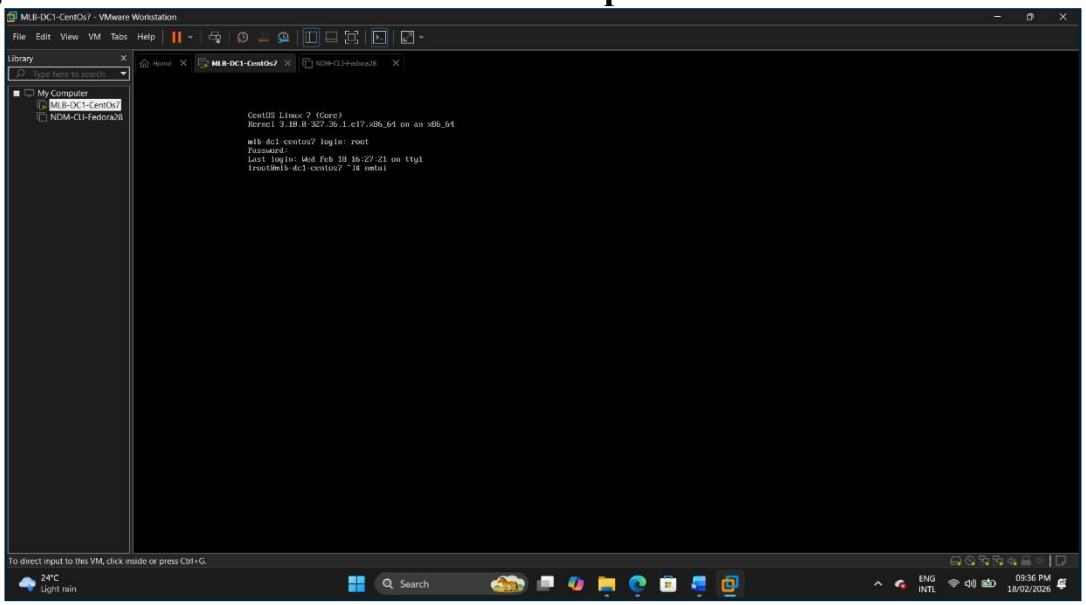
A client device makes a request to the DHCP server upon connecting to a network. The client is then given an available IP address by the server along with other settings like:

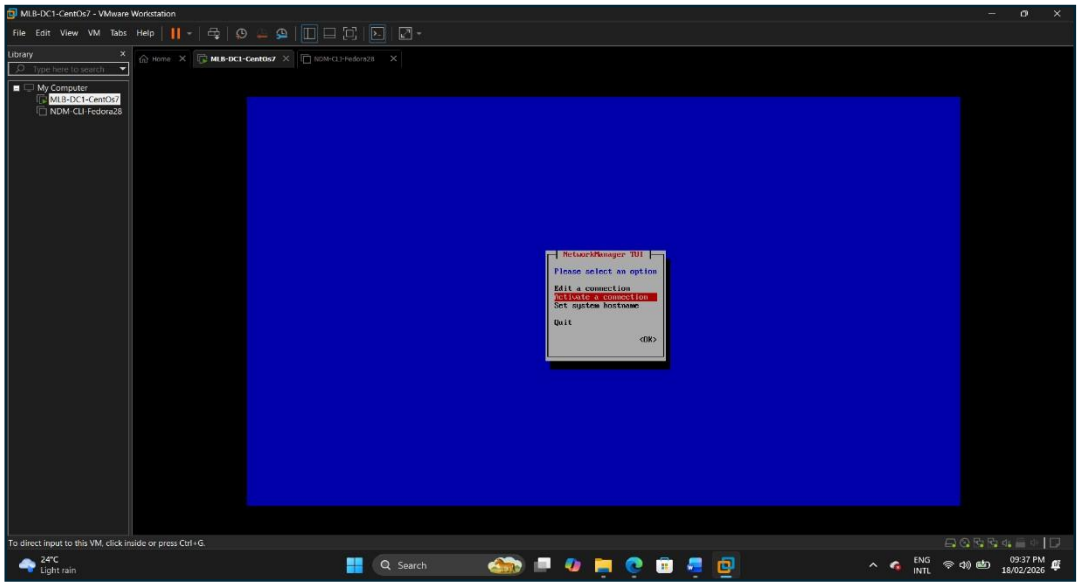
- Subnet mask
- Default gateway
- DNS server address

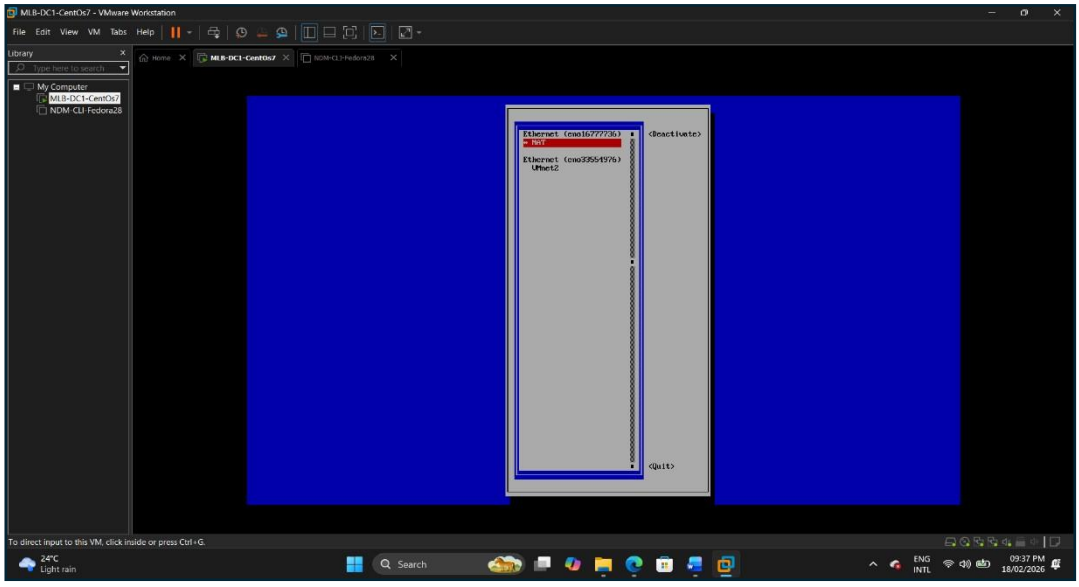
DHCP is widely used in enterprise networks Because DHCP makes IP address management easier, increases scalability, and lowers configuration errors, it is frequently utilized in home networks, business networks, and educational lab settings.

This lab shows how automatic IP configuration functions in a regulated network environment by installing and configuring DHCP on a CentOS server to dynamically assign IP addresses to a Fedora client computer.

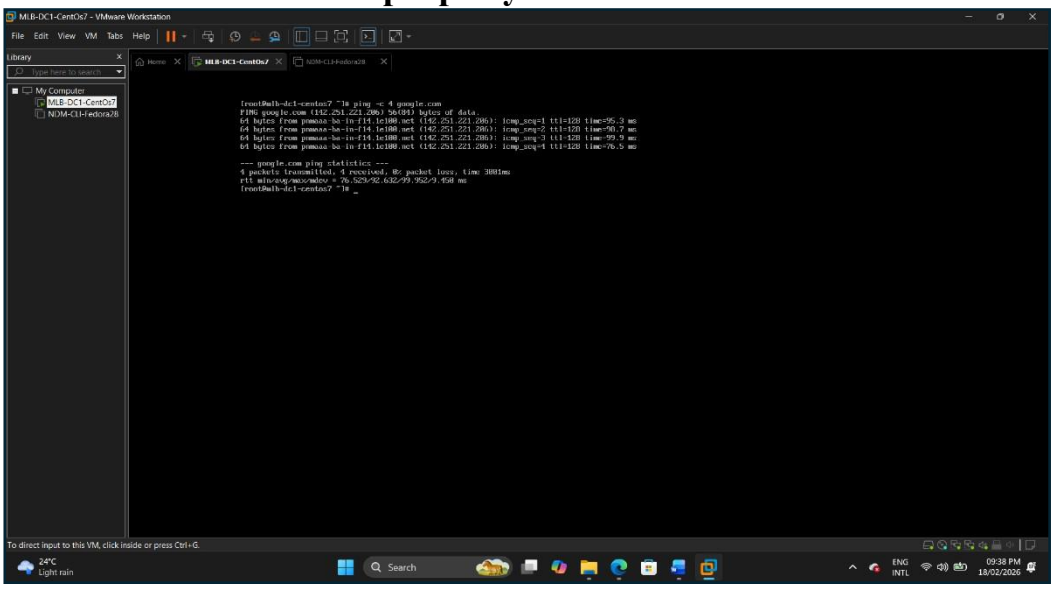
Login to CentOS and activate the “NAT” port





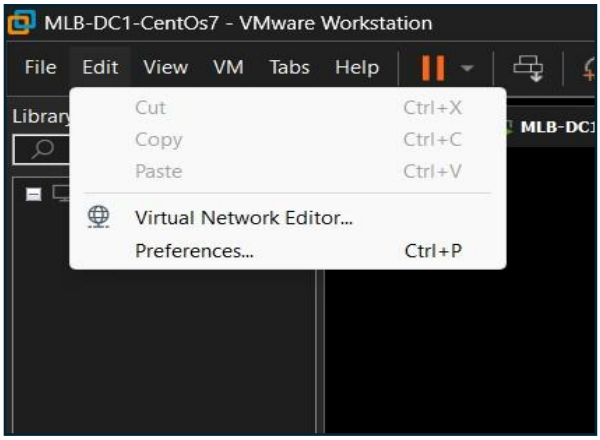


Check that the “NAT” work properly

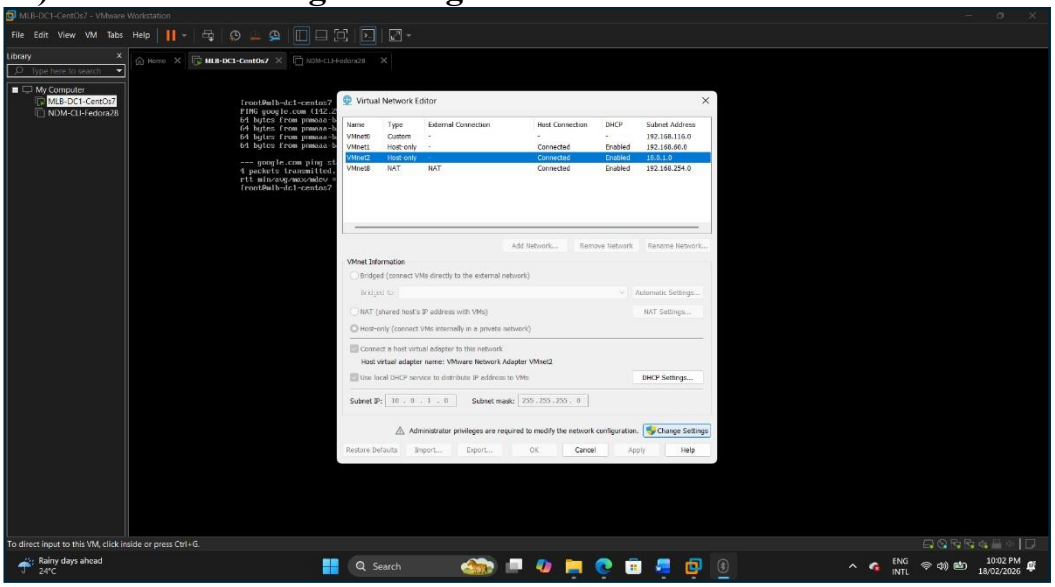


Disable the DHCP settings in VMnet 2

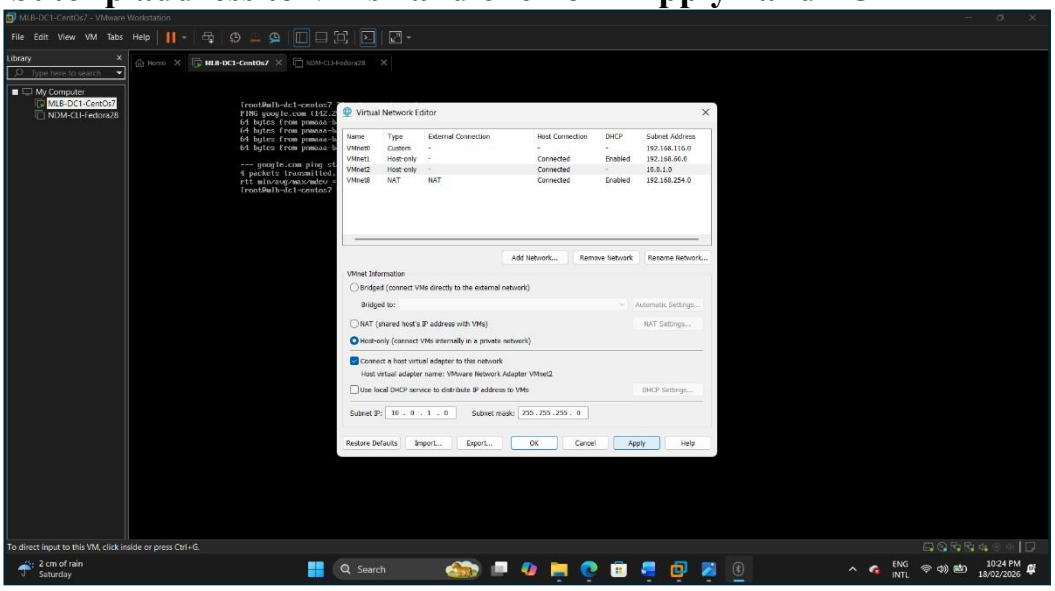
1) Go to “Virtual Network Editor” in “Edit” tab



2) Click on “Change Settings” button

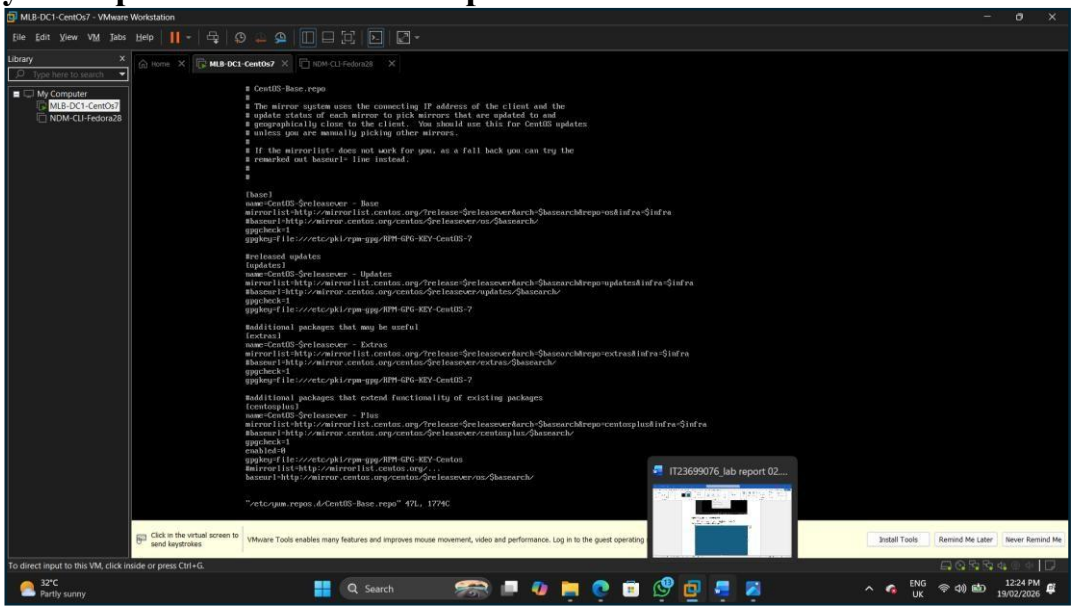


3) Select the “VMnet2” and untick the “Use local DHCP server to distribute Ip address to VMs” and click on “Apply” and “ÖK”

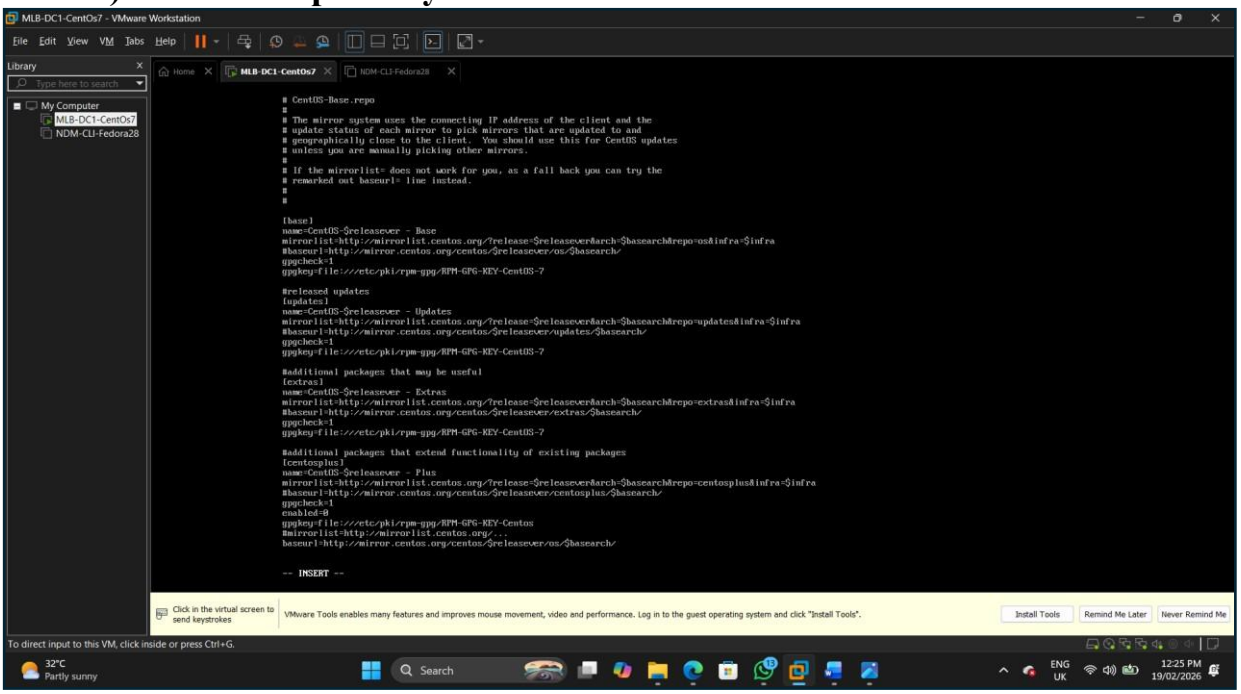


Update “baseurl” in Repository Files

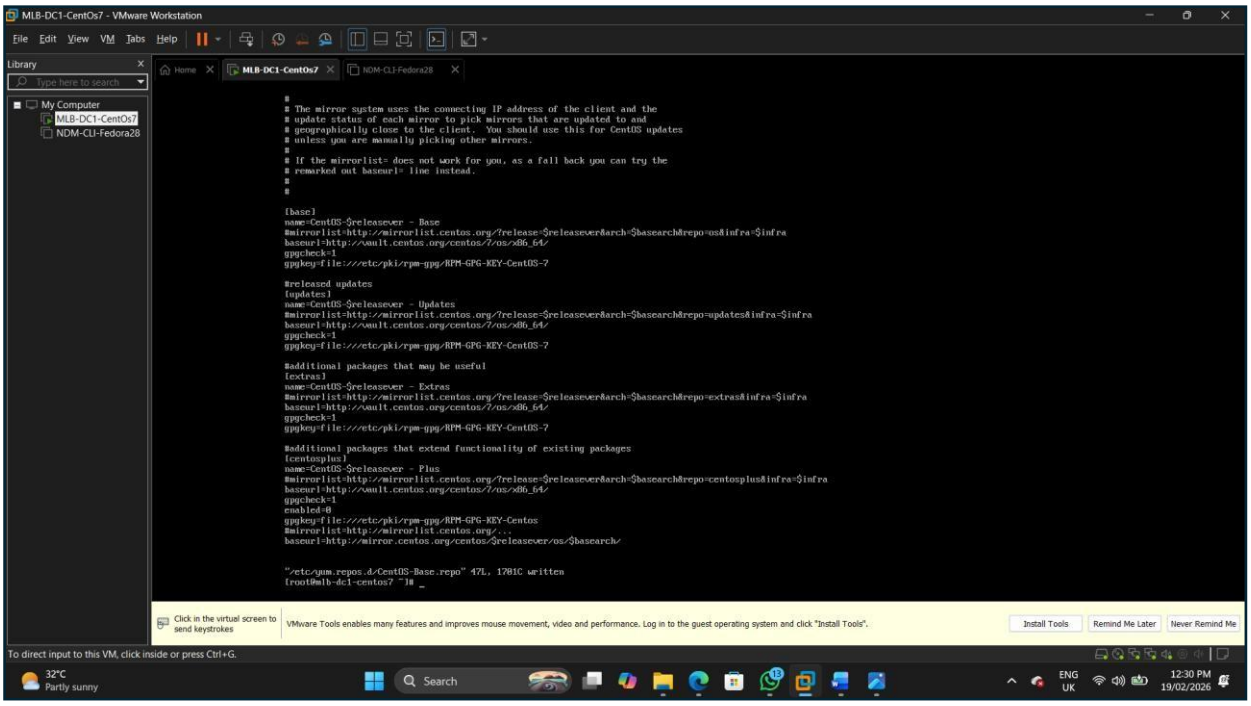
1) Open the repository configuration file using “vi /etc/yum.repos.d/CentOS-Base.repo”



2) Edit the Repository Files

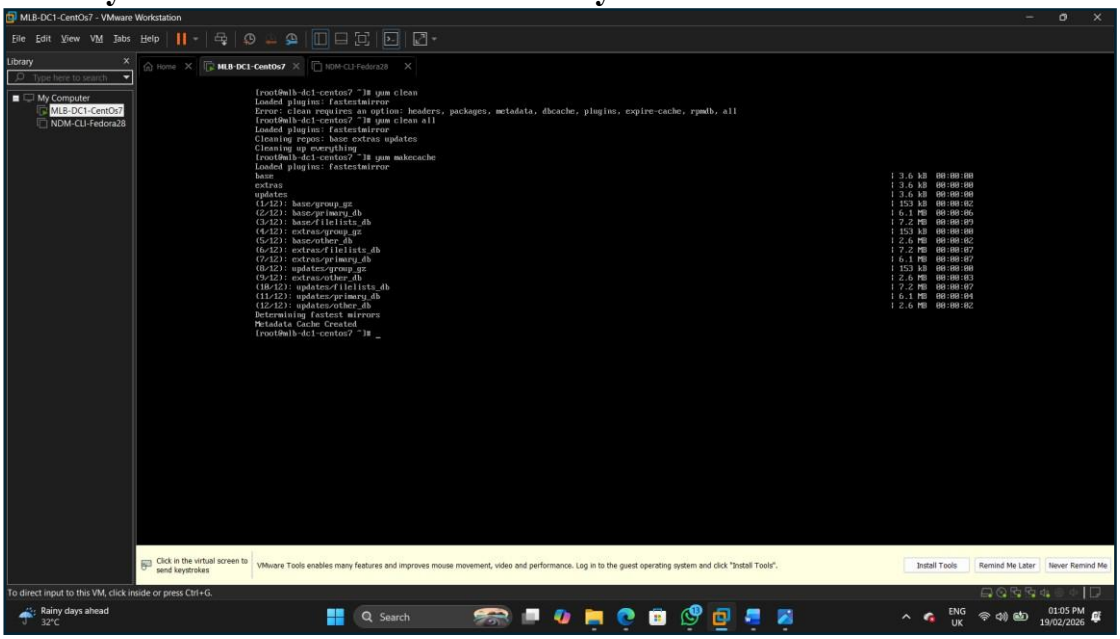


3) Save the edited Repository Files

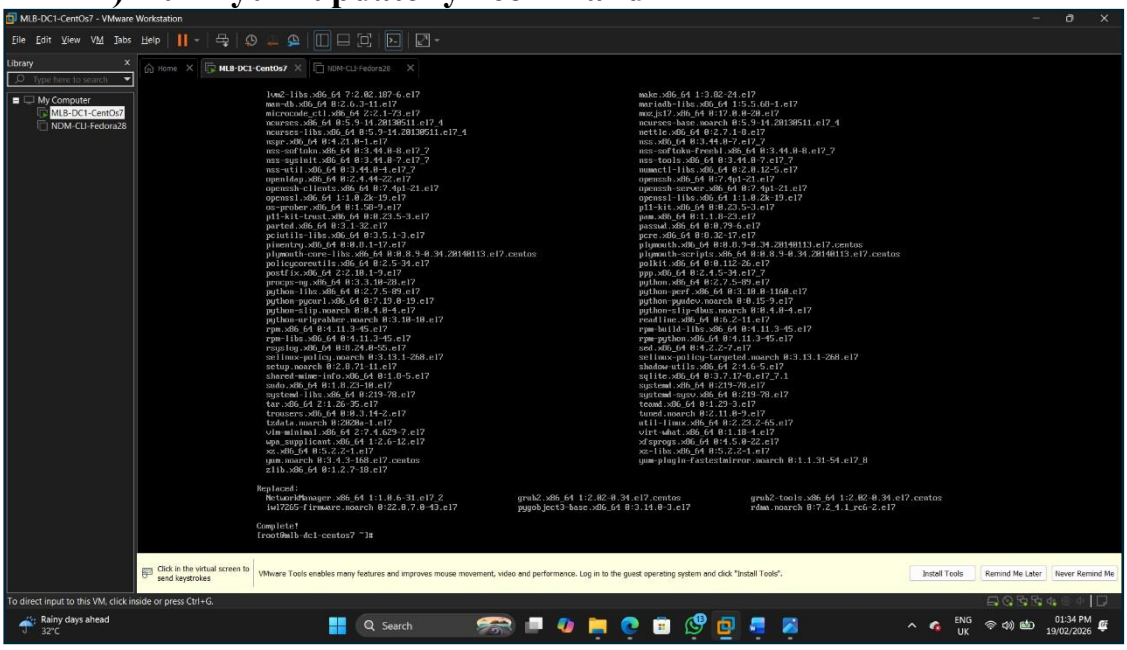


Clean and update the “yum”

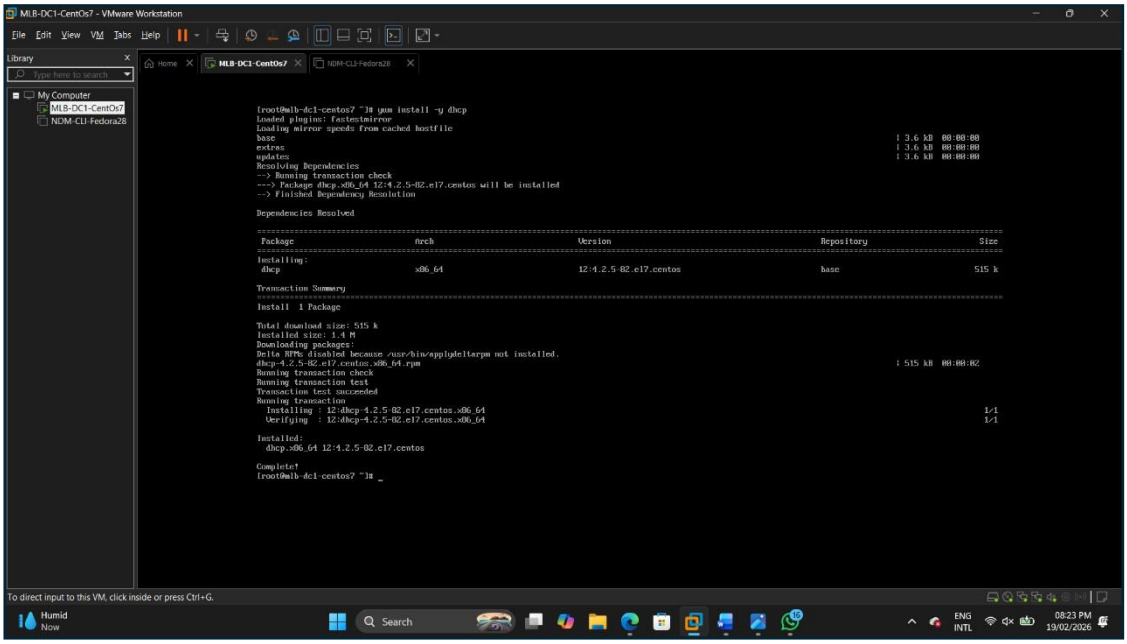
1) Run “yum clear all” command and “yum mackecashe” command



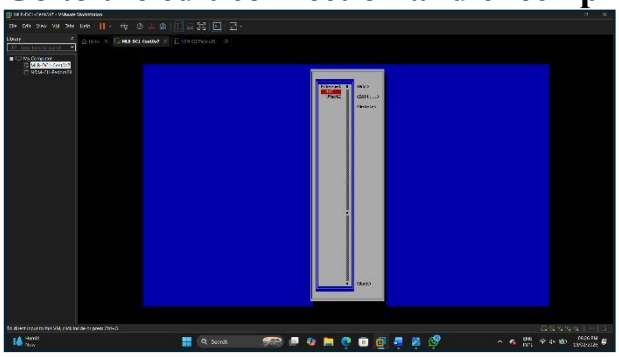
2) Run “yum update -y” command

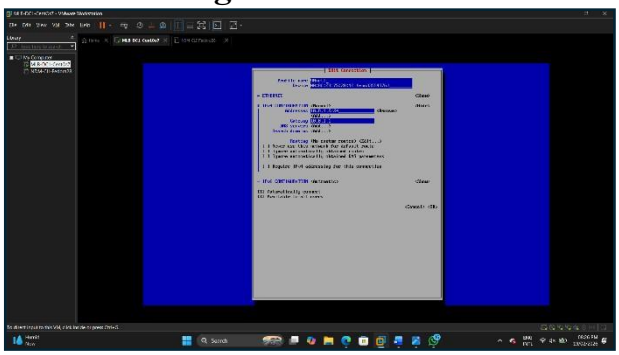


Install DHCP in CentOS

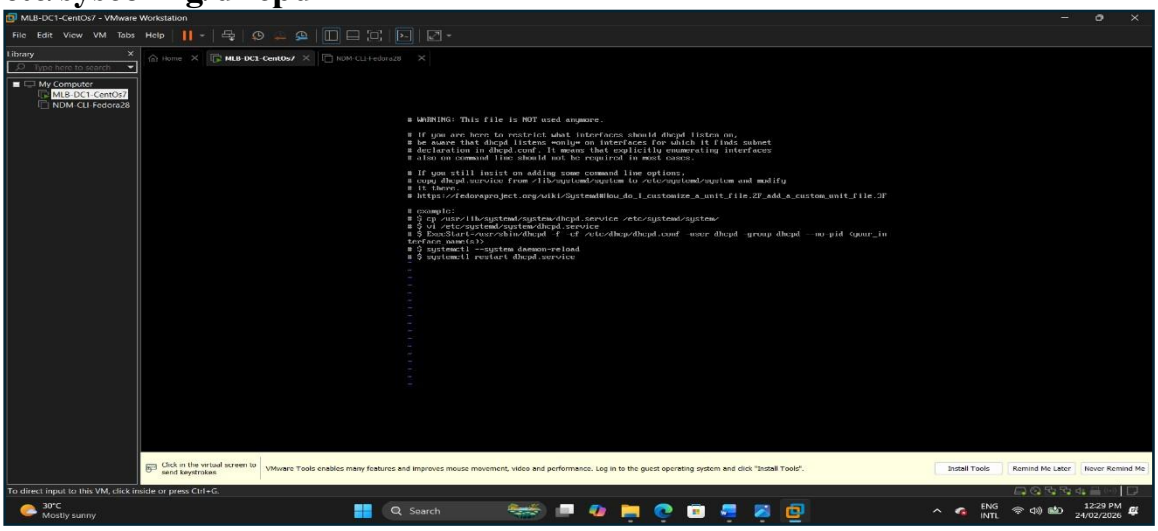


Go to the edit connection and check profile name using “nmtui”

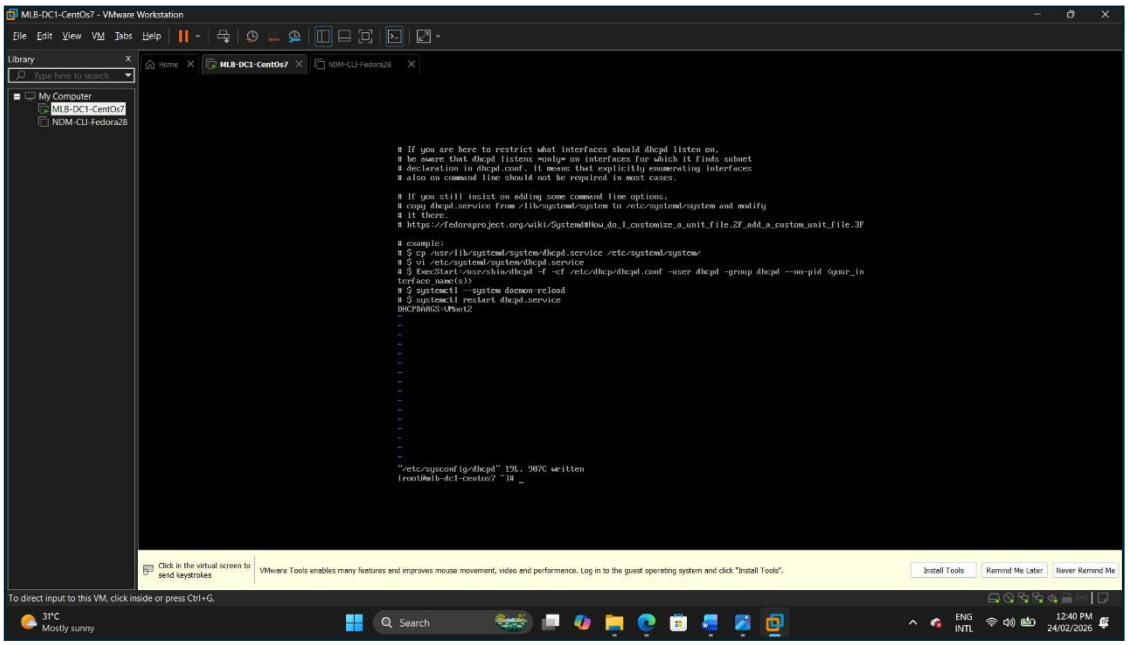




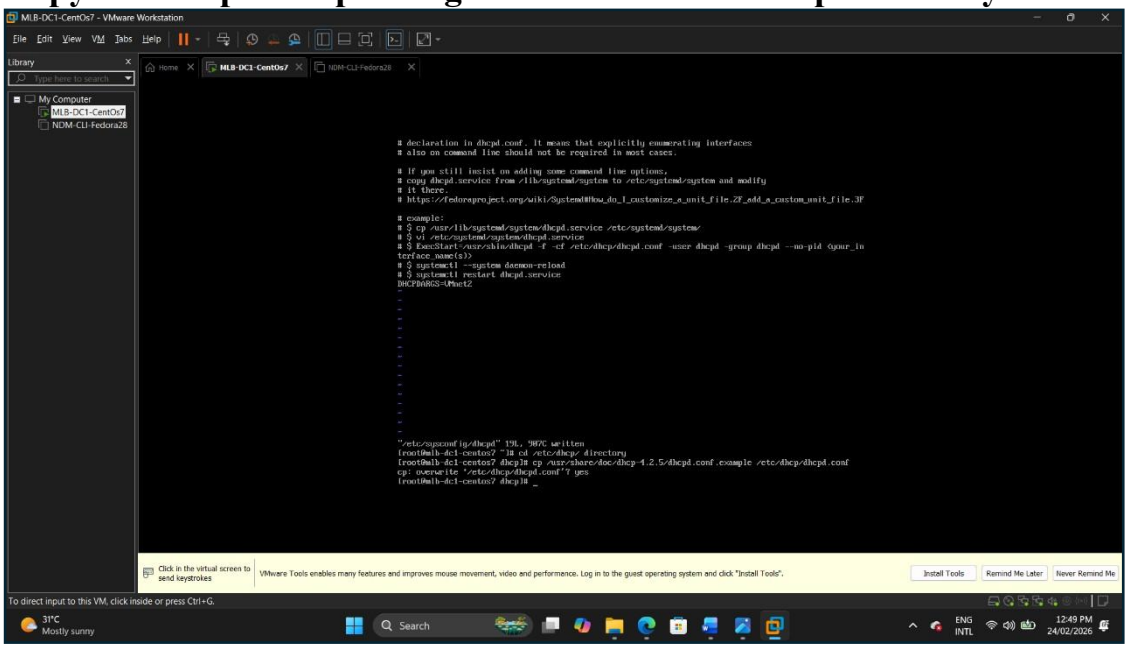
Configuring DHCP settings s assigning the network interface using “vi /etc/sysconfig/dhcpd”



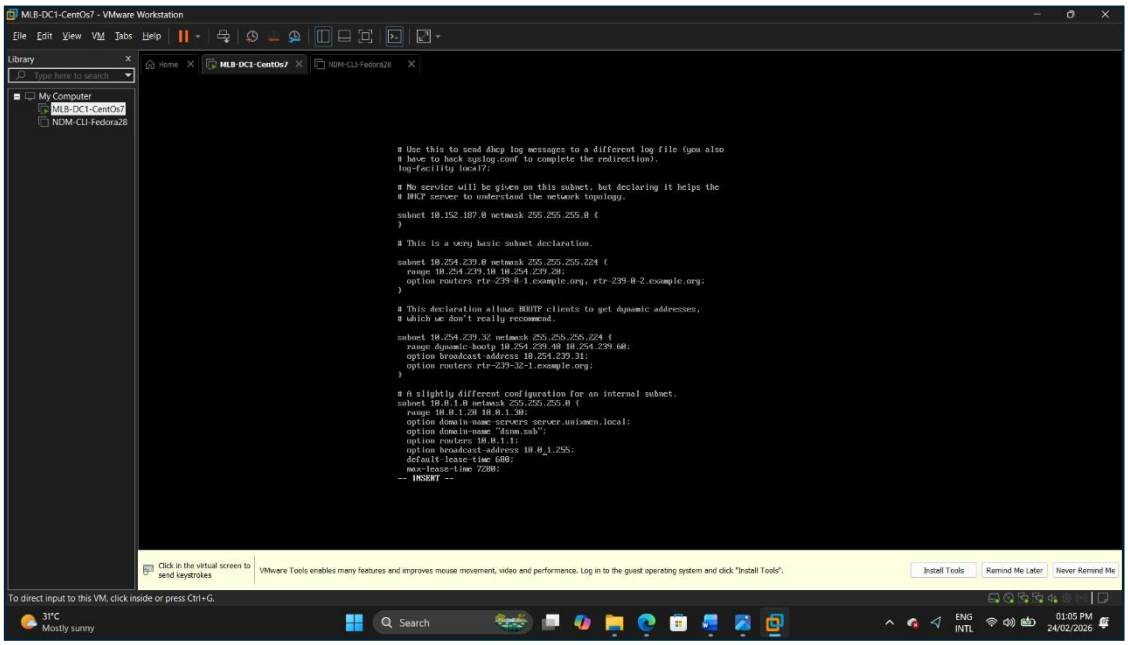
Insert the” vmnet2’s profile name in ‘nmtui’ using “DHCPDARGS” command and save



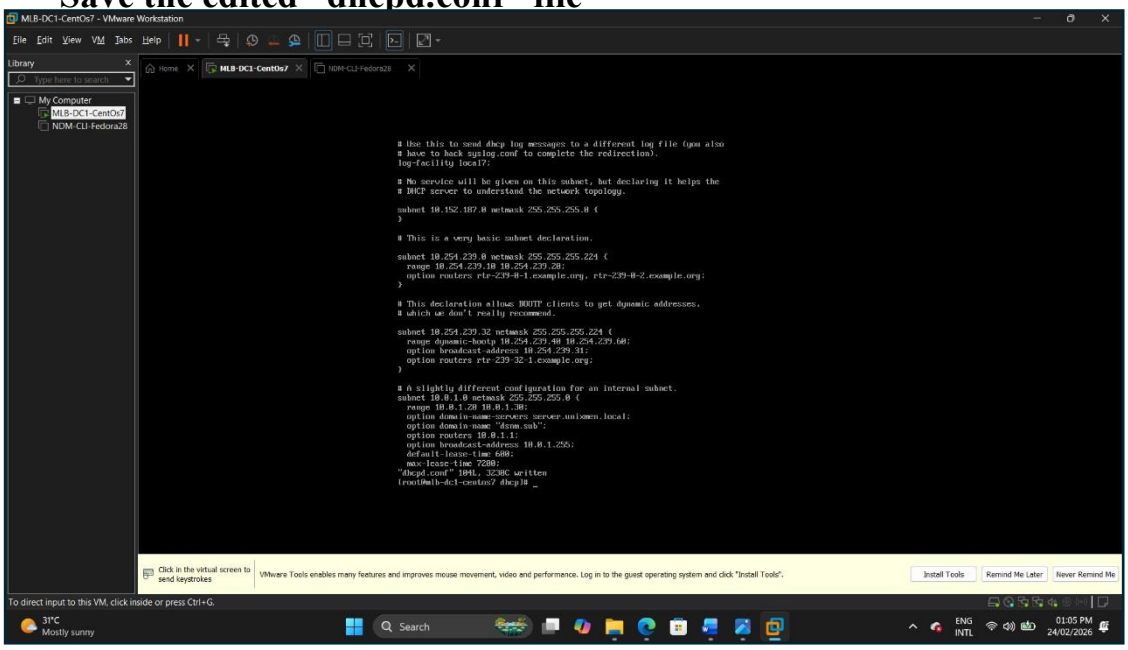
Copy the sample dhcp configuration file to /etc/dhcp/ directory



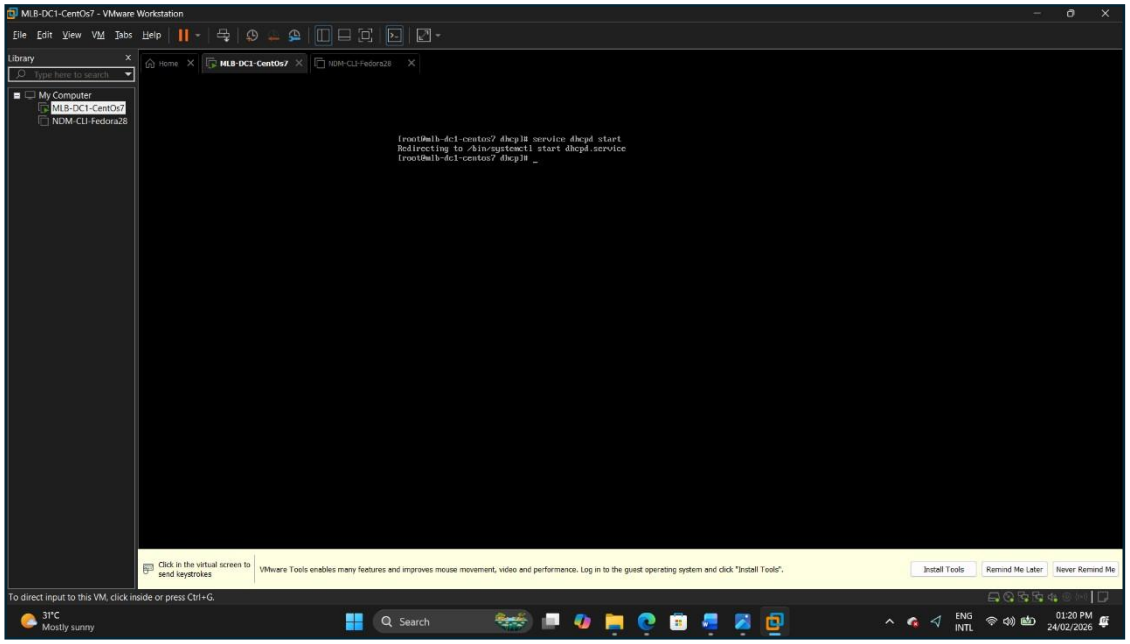
Open the “dhcpd.conf” file using “vi /etc/dhcp/dhcpd.conf” command and edit



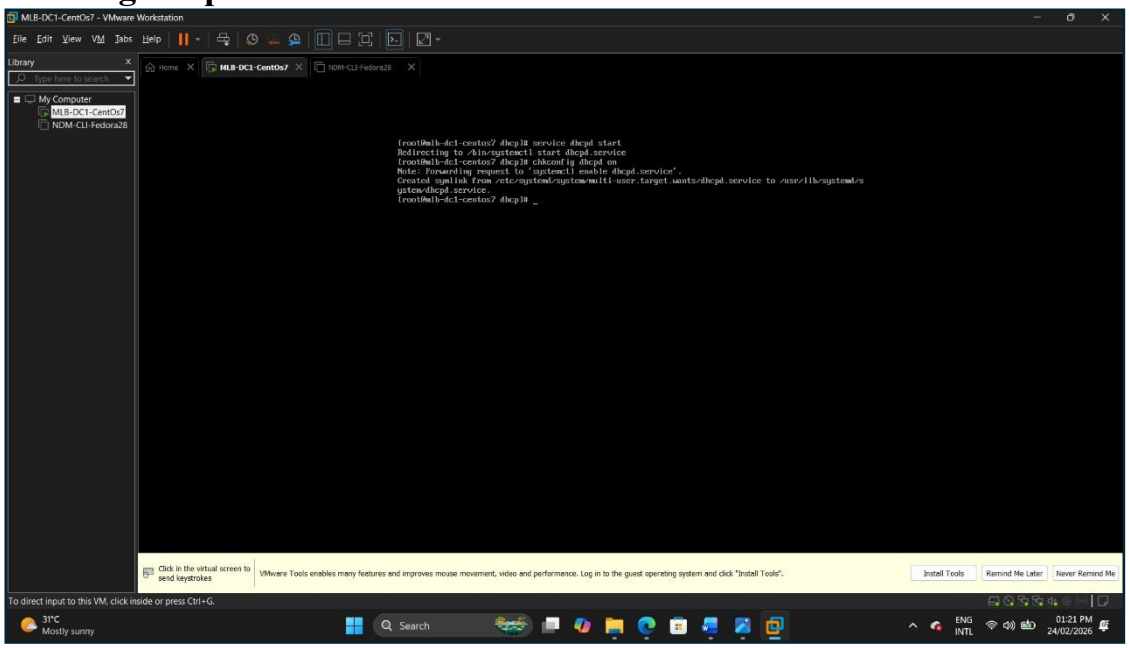
Save the edited “dhcpd.conf” file



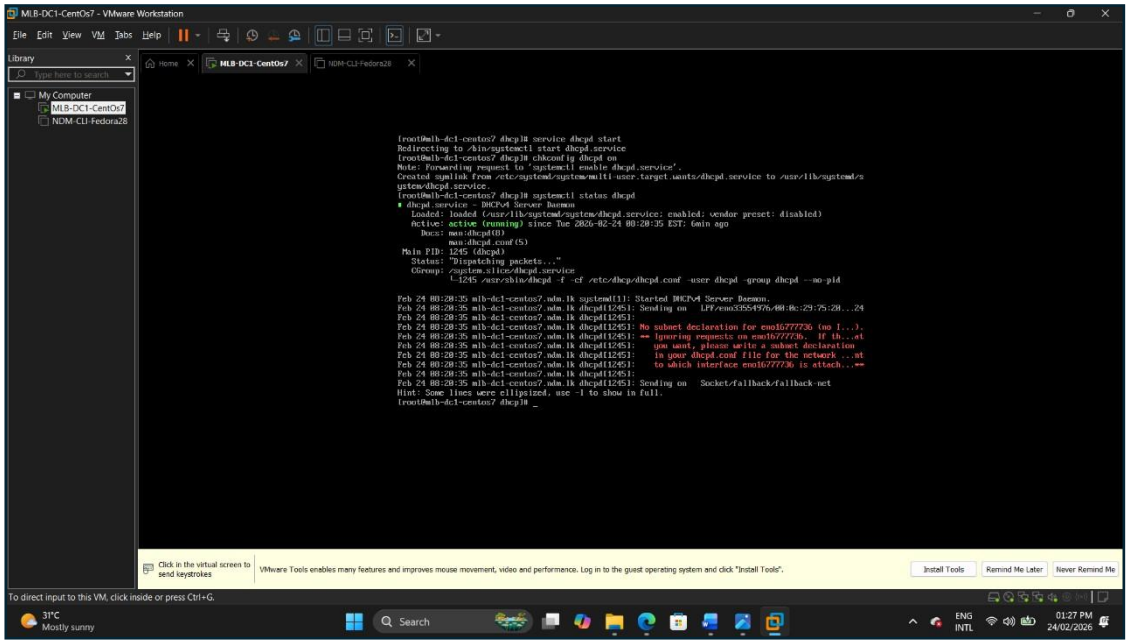
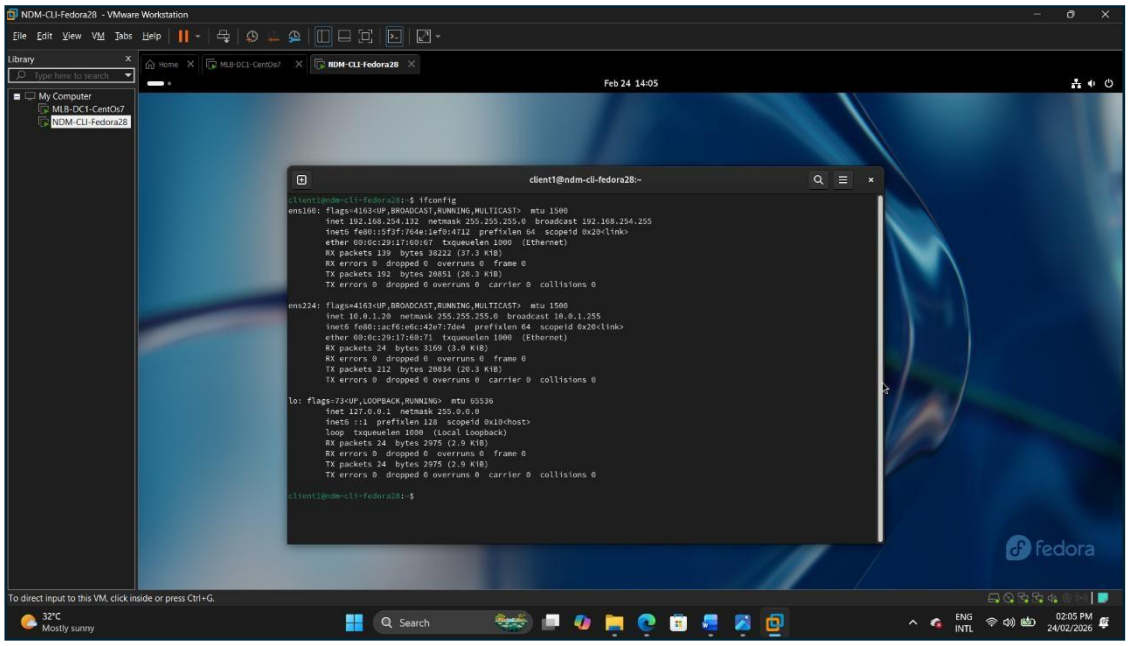
Start the DHCP service



Start up the DHCP server at logon to the server session using "chkconfig dhcpd on" command



Check the status of DHCP server and client



check the IP address of fedora and see if the DHCP server is working or not